



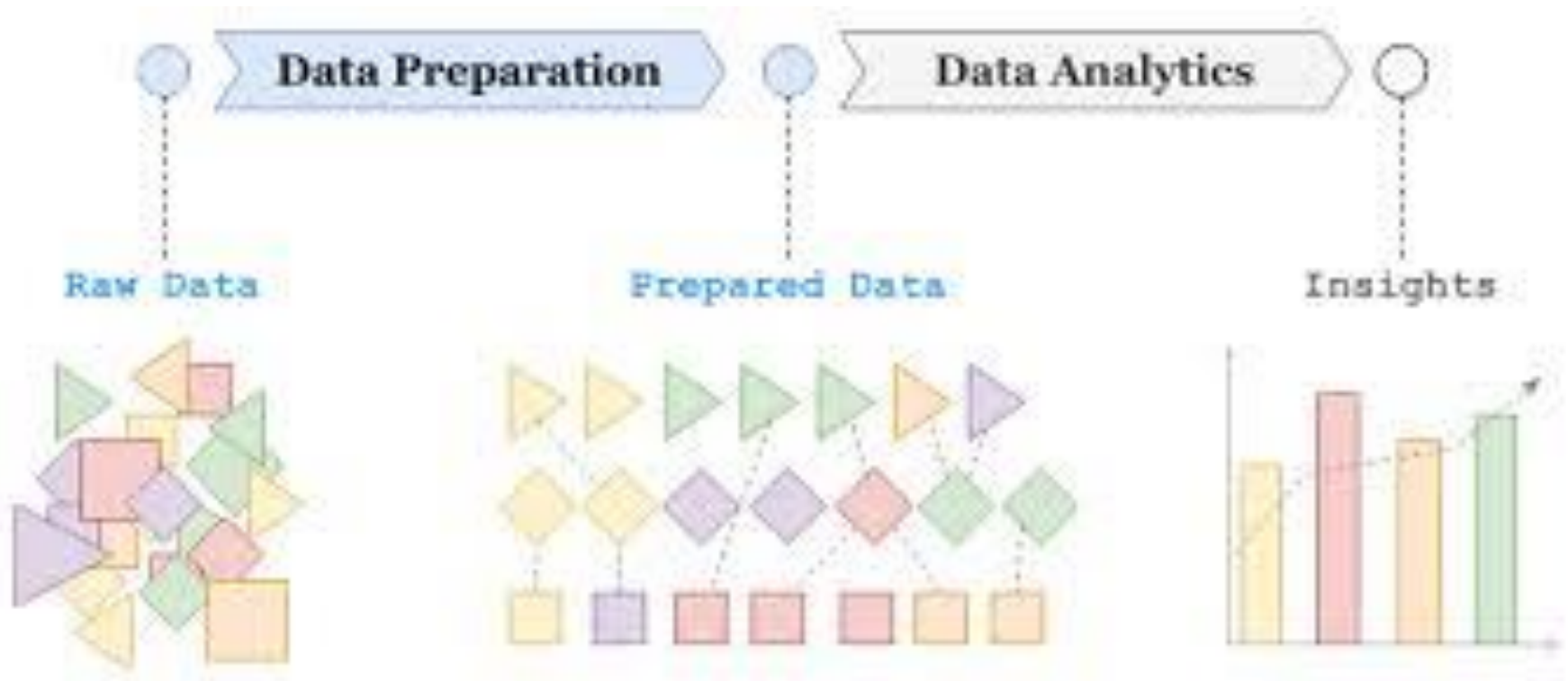
Data Preparation

- Data Preparation
 - Pre-Processing raw data to obtain a form suitable for analysis





Why to Prepare Data



To Analyze the data efficiently



Data Preparation Steps

1. Data Discretization

- Transforming data(values of attributes) in continuous domain to a finite set of values

2. Data Cleaning

- To handle incomplete, invalid and inconsistent data

3. Data Integration

- Combining data collected from various sources

4. Data Transformation

- Process of converting one format to another format for efficient analysis

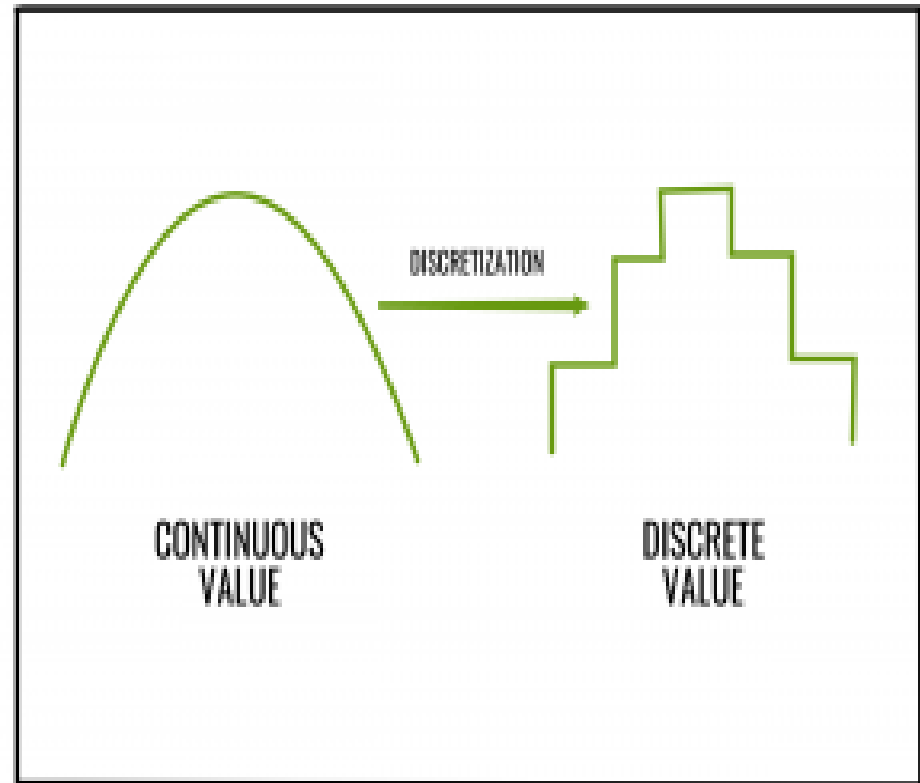
5. Data Reduction

- Reduced representation of data



Data Discretization

- 1) Binning
 - Equal Frequency binning
 - bins have an equal frequency
 - Equal width binning
 - bins have equal width with a range of each bin are defined as
[$\text{min} + w$, $\text{min} + 2w$]
[$\text{min} + nw$]
 - where $w = (\text{max} - \text{min}) / (\text{no of bins})$.





Data Discretization

- `df = pd.DataFrame({'rno': [4, 14, 17, 8, 12, 13, 15, 18, 22, 23, 24, 25],`
- `'cgpa': [9.2, 7.5, 9.4, 8.7, 6.7, 8.0,`
- `'feedback': [9, 7, 10, 6, 8, 8, 9, 9, 10,`
- `10, 8, 9]})`
- `print(df)`
- `df['cgpa_bin'] = pd.cut(df['cgpa'], bin=3)`
- `df['cgpa_bin'] = pd.qcut(df['cgpa'], q=3)`

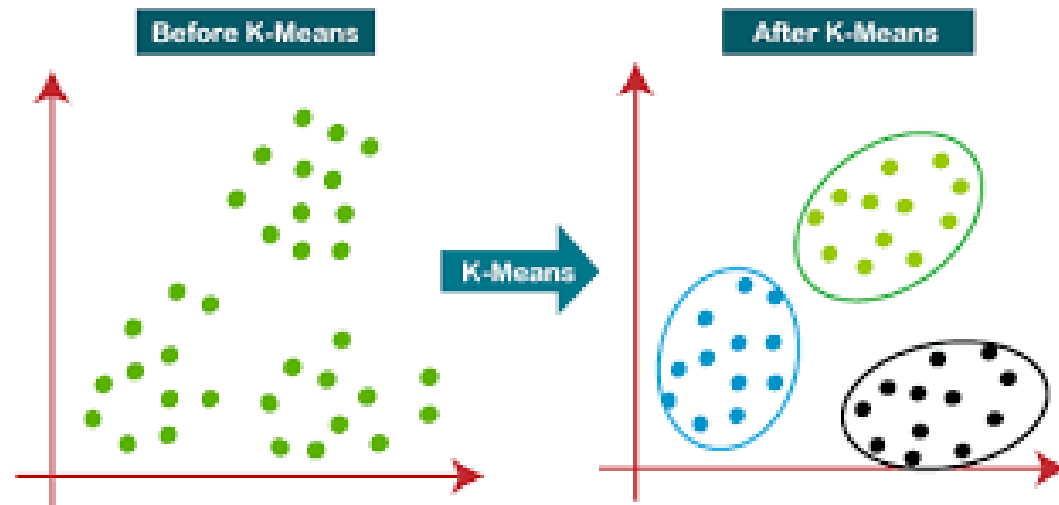
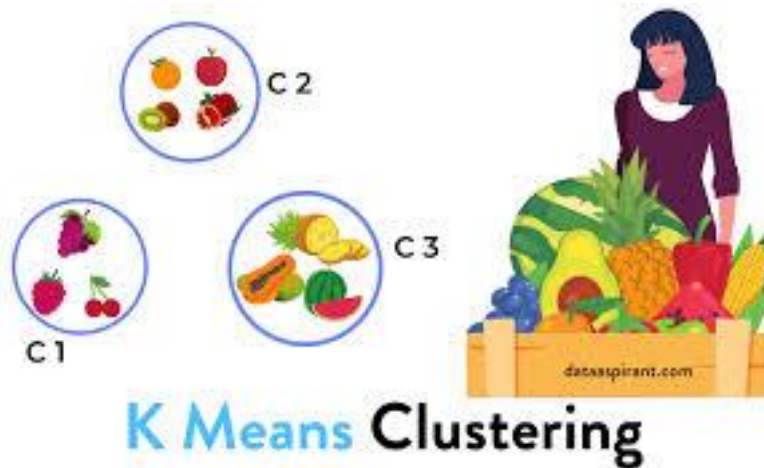
Data Discretization

- `df['cgpa_bin'].value_counts()`
- `df['cgpa_bin'] = pd.qcut(df['cgpa'], q=[0, .2, .4, .6, .8, 1])`
- `df['cgpa_bin'] = pd.qcut(df['cgpa'],`
- `q=[0, .2, .4, .6, .8, 1],`
- `labels=['A', 'B', 'C', 'D', 'E'])`
- `print(df)`



Data Discretization

- 2) Discretization Using Clustering
 - K Means Algorithm





Data Discretization

K-means algorithm Algorithm

Input:

$D = \{t_1, t_2, \dots, t_n\}$ // Set of elements

K // Number of desired clusters

Output:

K // Set of clusters

K-Means algorithm:

Assign initial values for $m_1, m_2, \dots, m_k$

repeat

assign each item t_i to the clusters which has the closest mean;

calculate new mean for each cluster;

until convergence criteria is met;



Data Discretization

